



Karnataka State Police
Government of Karnataka

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Dataathon 2026

Technology partner



Team Details

- a. **Team name:** Rangdari Gang
- b. **Team leader name:** Soma Mishra
- c. **Team size:** 2 (Sushanth Yelishetty and Soma Mishra)
- d. **Problem Statement:** AI-Driven Crime Analytics & Visualization Platform

Brief about the solution

Drishti links undetected cases by how the crime was done, not by who was named.

Karnataka carries thousands of undetected cases (final report type C). They sit in isolation because the FIR schema has **no global offender identity** — Accused.PersonID is documented as “A1, A2, A3...”, a sort order within a single FIR. When one offender works across several stations, the database holds several unrelated name strings and nobody ever joins them.

For every undetected FIR the engine builds a **behavioural fingerprint** from seven signals — the modus operandi in the complaint text, offence type, the Act and Section invoked, location, time-of-day pattern and victim profile. It scores every pair of cases against each other, then groups the look-alikes using **agglomerative clustering** on the resulting distance matrix.

The output is not a map of where crime happened. It is a **ranked queue of suspected offender groups**, each citing the real FIR numbers that make it up, each showing which signals produced the link. From **789 undetected cases across 69 stations and 19 districts**, it surfaces **16 groups holding 87 cases** — **every one of them spanning more than one police station.**

Opportunities

a. How different is it from any of the other existing ideas?

Typical crime tools

→ Drishti

- Descriptive dashboards showing where crime happened → **Prescriptive — surfaces hidden links and actionable case series to reopen**
- Depend on offender IDs or manual case linking → **Works without any person ID, using behavioural fingerprinting**
- Black-box predictions with limited transparency → **Explainable evidence trail with measured performance — P 0.928 / R 0.965 / F1 0.946**
- Perform poorly on sparse free-text reports → **Degrades gracefully — weights renormalise onto typology and spatiotemporal signals**
- Rely on external LLM APIs, with per-query cost and off-premises processing → **On-premises friendly — deterministic scikit-learn, no LLM API, data stays in Catalyst**

b. How will it be able to solve the problem?

An investigating officer sees their own station's cases. Nobody is positioned to notice that four robberies in four different stations share a method, an hour and a radius. Drishti runs that comparison statewide and returns a ranked list of groups worth reopening, with the evidence attached so the officer can judge it rather than trust it. It then closes the loop forward: a newly registered FIR is scored against every known group as it lands, so a case joins a pattern **before** it is worked as an isolated file.

c. USP of the proposed solution

Works without an offender ID — linkage is behavioural, so it runs on the existing FIR schema unchanged.

Every claim is evidence-backed — per-signal contributions and cited CrimeNo values. If the engine cannot cite it, it does not say it.

Accuracy measured inside the product — precision 0.928, recall 0.965, F1 0.946, computed live by the same engine that serves the app.

Identity carries zero weight, and we proved that was right — letting the accused name influence linkage made the engine measurably worse, so we removed it.

List of features offered by the solution

Undetected case linkage — behavioural fingerprint over seven weighted signals, composite pairwise similarity, agglomerative clustering into ranked offender groups

Evidence trail on every link — per-signal contribution breakdown, cited FIR numbers, spatial span and a full case timeline

Next-strike projection — time window, area radius and usual hours from the group's median inter-event gap and 90th-percentile spread, with a stated confidence

Live FIR triage — an incoming case scored against every known group and graded against that group's own cohesion — Escalate, Review, or treat as unlinked

Persons of interest — cross-FIR name resolution with fuzzy matching for transliteration drift

Interactive 3D district map — WebGL terrain extruded from real district boundaries, with case pins, linked-group webs and the projected zone

Natural-language search — retrieval-only, so every answer cites real records and cannot fabricate

Printable case brief — server-rendered PDF dossier with evidence, timeline, forecast and signature block

Role-based access control — investigator, analyst and supervisor, enforced at the API gateway; investigators scoped to their district

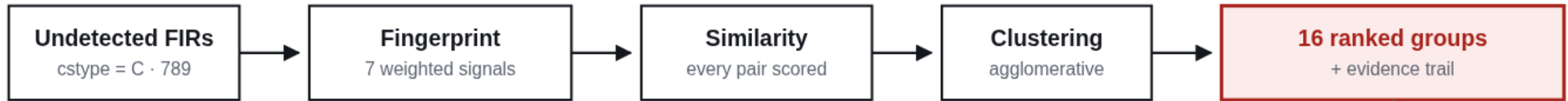
In-product accuracy panel — precision, recall, F1, per-signal ablation and a written statement of where the method fails

Bilingual interface — English and Kannada across all labels and explanations

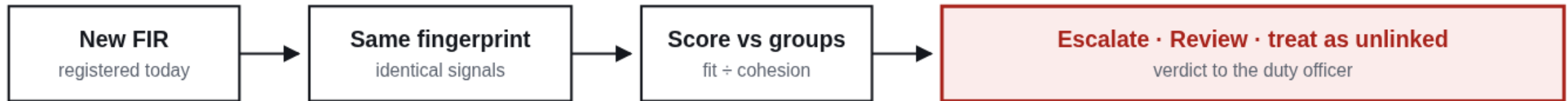


Process flow diagram

ARCHIVE PASS · RECOMPUTED NIGHTLY



LIVE INTAKE · PER NEW FIR



The same fingerprint runs in both passes — that is what lets a brand-new FIR be judged against groups built from the archive, and why the two lanes are one system rather than two.

Archive pass — every undetected FIR is fingerprinted on seven weighted signals, scored against every other case, and clustered. Groups are then ranked by cohesion multiplied by actionability, which favours jurisdictional reach, recency and gravity. Runs nightly.

Live intake — a newly registered FIR is fingerprinted identically and scored against each group, then graded against that group's own internal cohesion rather than a fixed threshold. A tight group demands a high score; a loose one cannot reach it.

Nothing is asserted. Each group carries the per-signal contributions that produced it and the real FIR numbers it cites, so an investigator can audit the reasoning rather than trust the output.



Wireframes / Mock diagrams of the proposed solution

Command bar

retrieval-only search;
every answer cites a real FIR

Primary navigation

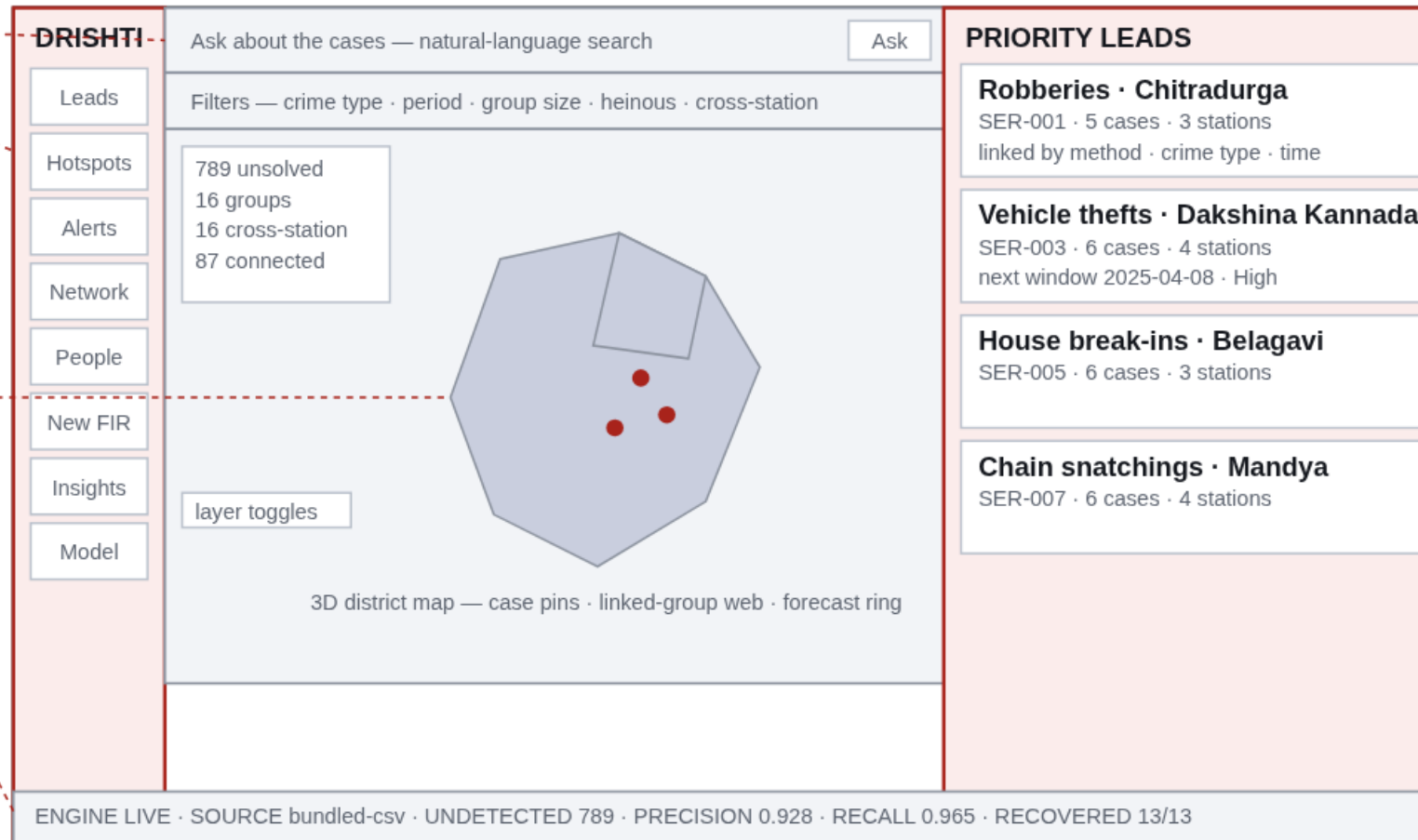
eight analytical views;
locked by role

Map stage

districts extruded by
case volume; click to
drop case pins

Status bar

live engine state and
measured accuracy



Dossier panel

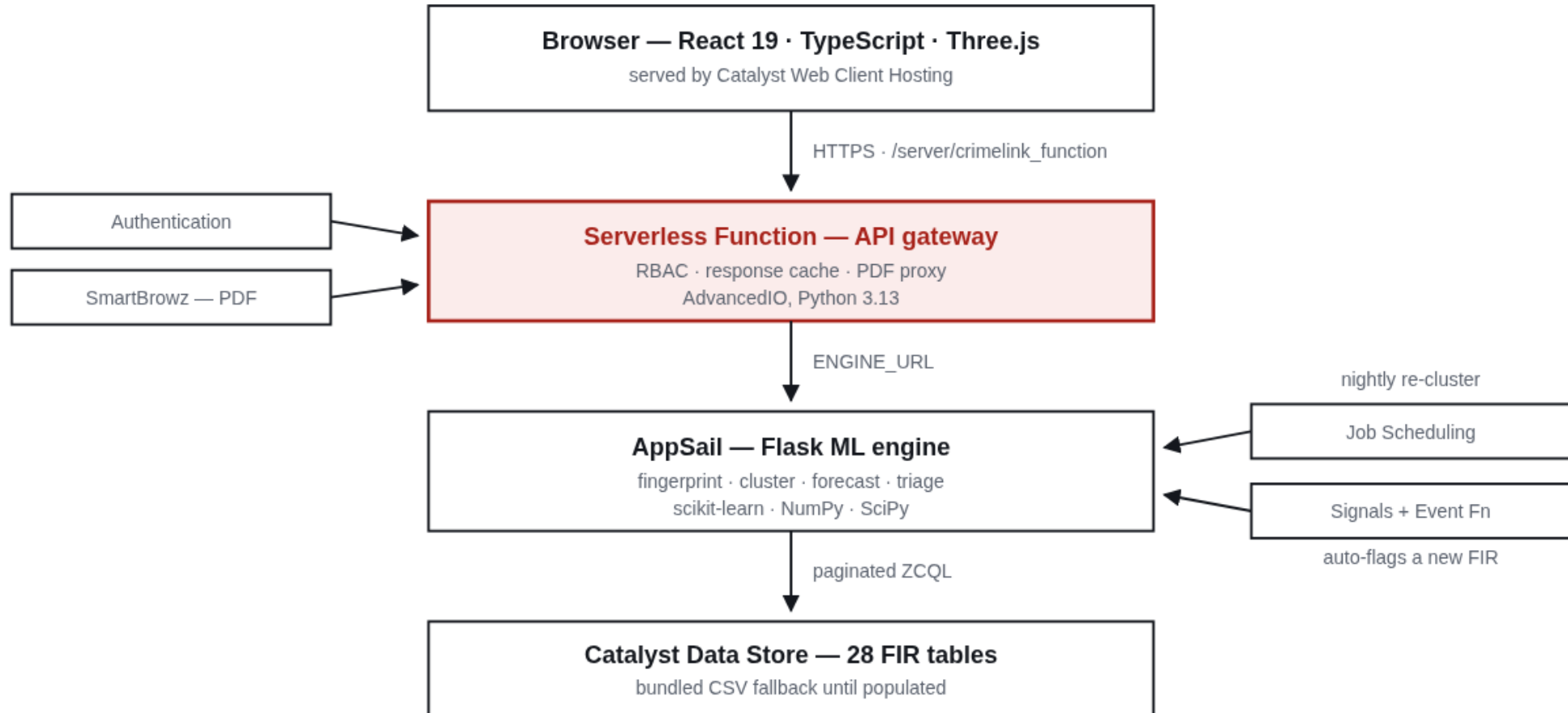
ranked leads; opens
to a full case file

Evidence on every card

why it linked, and
the projected window

Three regions, each doing one job: navigation on the rail, the map as the workspace, and the dossier where a lead opens into a full case file with its evidence. Live screens of this layout follow on the next slides.

Architecture diagram of the proposed solution



Access control sits at the gateway, so no request reaches case data before a role is resolved. The engine is the only component that touches the Data Store.

Technologies to be used in the solution

Frontend	React 19, TypeScript, Three.js with React-Three-Fiber (WebGL 3D map)
Machine learning	Python 3.13, scikit-learn (TF-IDF, AgglomerativeClustering), NumPy, SciPy, NetworkX
API layer	Flask on AppSail; Catalyst Serverless Function (AdvancedIO) as the gateway
Document generation	fpdf2 for the server-rendered case brief; SmartBrowz as the Catalyst-native renderer
Data	Catalyst Data Store queried with ZCQL; schema-faithful CSV fallback
Geospatial	Karnataka district boundary GeoJSON, Douglas-Peucker simplification, haversine distance

Techniques

Behavioural fingerprinting · TF-IDF over complaint text · weighted composite similarity with per-pair weight renormalisation · agglomerative clustering on a precomputed distance matrix · leave-one-signal-out ablation · hold-one-out forecast back-testing · Douglas-Peucker simplification for the district meshes.

Design constraints we held to

- Clustering stays on **scikit-learn primitives** on purpose, so the whole engine installs from requirements.txt on the Catalyst managed runtime with **no native compilation**.

No external LLM API. Retrieval is deterministic, so no case data leaves the Catalyst environment and there is no per-query cost.

CPU only — no GPU anywhere in the pipeline.

Catalyst Services being used in the solution

AppSail	Hosts the ML engine — fingerprinting, clustering, forecast, triage, PDF
Serverless Functions	API gateway; enforces role-based access before any case data is read
Data Store	The FIR schema — 28 tables, queried by ZCQL
Authentication	Identity behind the investigator / analyst / supervisor roles
Cache	Caches computed API responses across function instances
Job Scheduling	Nightly re-cluster and cache flush — used instead of Cron, which is deprecated
Signals + Event Functions	Auto-triages a newly registered FIR — instead of Event Listeners, also deprecated
SmartBrowz	Renders the case-brief PDF
Web Client Hosting	Serves the React application

We deliberately avoided the deprecated services: **Job Scheduling** replaces **Cron** and **Signals with Event Functions** replaces **Event Listeners**, both end-of-life 30 April 2026.

Since the previous submission, four services moved from roadmap to shipped — Authentication (role-based access control), Job Scheduling, Signals with Event Functions, and SmartBrowz. Data Store full-text search on BriefFacts activates when the tables are populated.

Estimated implementation cost

No new infrastructure and no new data collection. Costs scale gradually with usage.

Pilot / demo stage — runs on the Catalyst Developer Tier at roughly ₹0, within the free tier.

District-level deployment ($\sim 10^4$ – 10^5 FIRs) — AppSail, Serverless Functions and Data Store, in the **low thousands of rupees per month**.

State-wide ($\sim 10^6$ FIRs with nightly recomputation) — **tens of thousands of rupees per month**, driven mainly by AppSail and Data Store scaling.

Costs stay low because the pipeline uses **CPU-based scikit-learn models** — **no GPUs and no external LLM API calls** — and linkage runs as a nightly batch rather than on every user query. Figures are estimates and vary with data volume and recomputation frequency.

What is not in the cost

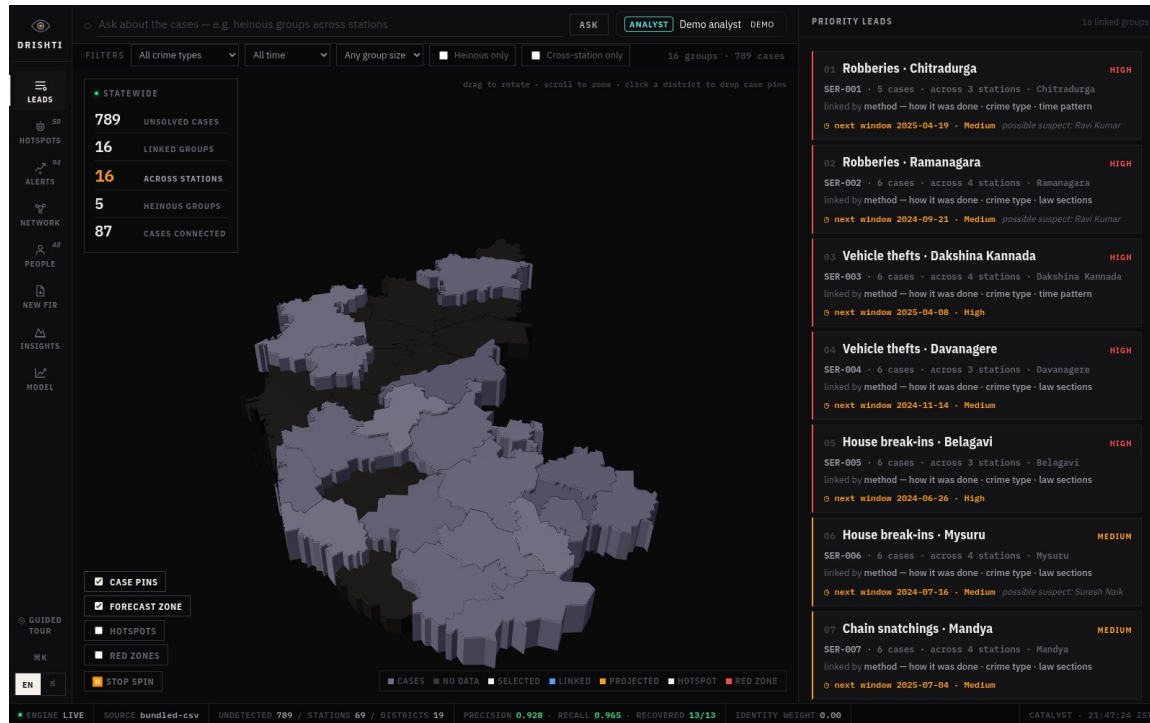
No new data collection — it reads the FIR schema exactly as it exists.

No change to officer workflow — no additional field, form or step when filing an FIR.

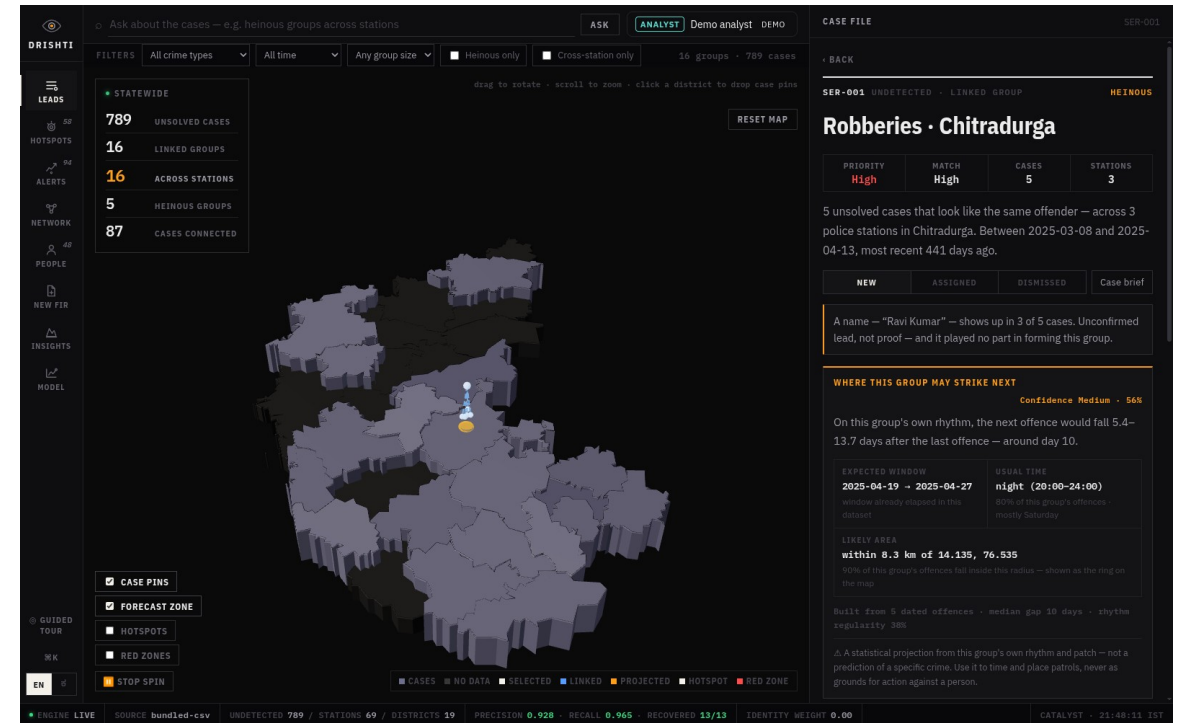
No third-party licences — every component is either open-source or a Catalyst service already in the KSP stack.

No GPU or inference hardware, and no per-query API spend that grows with usage.

Snapshots of the prototype

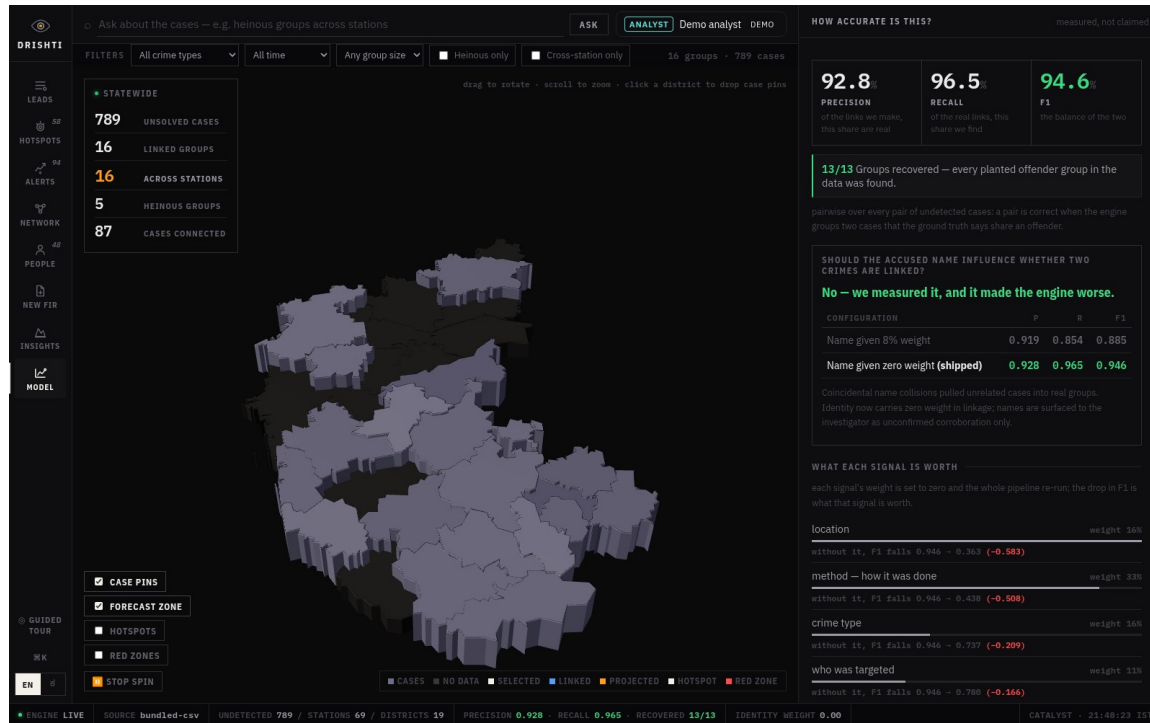


Statewide — 789 undetected cases, 16 linked groups

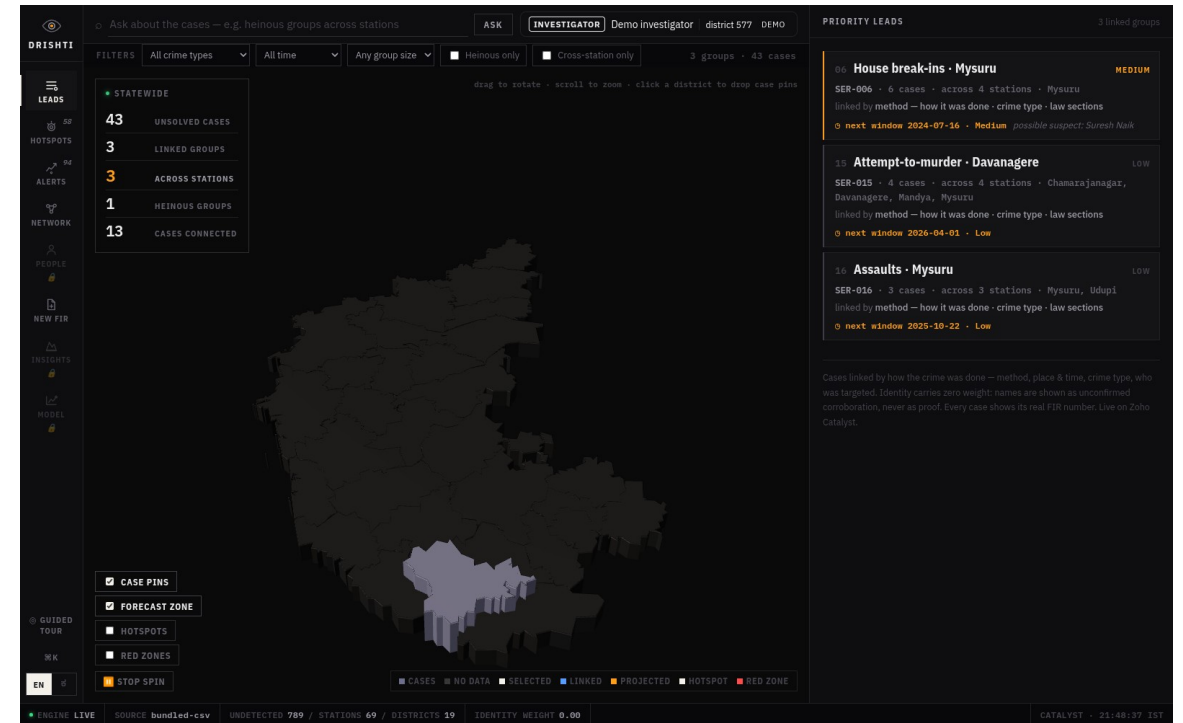


A linked group — evidence trail and next-strike projection

Snapshots of the prototype



Measured accuracy, ablation and the identity experiment



Role-based access — an investigator sees one district



Prototype Performance report / Benchmarking

Linkage accuracy — measured pairwise over every pair of undetected cases against planted ground truth

Precision 0.928 · Recall 0.965 · F1 0.946 · 13 of 13 planted groups recovered. Reproducible offline in one command: `python appsail-python/run_local.py`

What each signal is worth (leave-one-out ablation)

Location removed → F1 falls 0.946 to **0.363**. Method (MO text) removed → **0.438**. Crime type removed → **0.737**.

The identity experiment

Accused name at 8% weight → P 0.919 / R 0.854 / F1 0.885. **At zero weight (shipped) → P 0.928 / R 0.965 / F1 0.946.** Coincidental name collisions pulled unrelated cases into real groups, so identity now carries no weight in linkage at all.

Forecast, back-tested

Each group's last offence held out, a projection built from the earlier events only, then checked against what actually happened: **area hit rate 92.9%**, window hit rate 50.0%, median timing error 7.7 days, across 14 groups. The area holds up far better than the timing — which is why the map ring is the actionable part of the forecast, not the date.

Schema conformance

28 of 28 ER tables implemented, 42 of 42 foreign keys resolve. The KSP dataset was not released, so these figures are measured on schema-faithful synthetic data with planted offender groups. Real FIR data swaps in through the same tables and pipeline unchanged, and the accuracy panel re-measures itself against it.

Provide links to your:

1. GitHub Public Repository

<https://github.com/whysush/drishti-crimelink>

2. Demo Video Link (3 minutes)

<https://drive.google.com/file/d/1-NFxiHr9veflHYR7-AuijxACmk5u-CWV/view?usp=sharing>

3. Deployed Link

<https://crimelink-927199644.development.catalystserverless.com/app/index.html>

Live on Zoho Catalyst — React client on Web Client Hosting, Serverless Function gateway, and the ML engine on AppSail.

What is in each

Repository — the full linkage engine, forecast, triage and person resolution; the API gateway with its access-control layer; the job and event functions; the React client; and the synthetic data generator with its ground truth, so every figure on slide 12 can be reproduced.

Deployed link — open it a minute or two before viewing. The engine idles between uses and the status bar reads ENGINE LIVE once it is warm.

Deep links worth trying — ?tab=model for measured accuracy, ?group=SER-001 for a linked group with its forecast, and ?role=investigator&district=577 to see the access model restrict the view.

Additional Details / Future Development

Go live on real FIR data. The Data Store schema and loader are built and the engine already has a Data Store backend; populating the 28 tables switches the source with no code change. Full-text search on BriefFacts comes with it.

Promote to a Production environment. The prototype runs on the Catalyst Development environment.

Close the loop on the auto-flag. The Signals event function is deployed; binding it to the FIR insert event turns triage from a screen an officer opens into something that happens the moment a case is registered.

Kannada voice and translation. The interface is already bilingual; Zia Services would add speech input and translation of free-text complaints.

QuickML for richer semantic search. Retrieval today is deterministic; a RAG layer over the same cited records would widen natural-language querying without loosening the no-hallucination guarantee.

Zia AutoML for district-level risk forecasting, complementing the per-group projection already shipped.

Learn from investigator feedback. Every group can be marked assigned or dismissed — those judgements are the labels needed to tune signal weights against real outcomes rather than planted ground truth.

Where the method fails — stated on the slide deliberately

Links are **inferred, never asserted** — a lead to investigate, not evidence. Two unrelated offenders working the same market at the same hour can look alike. Sparse complaint text weakens the strongest signal, though the weights renormalise so linkage still runs. Complainant caste and religion exist in the schema and are **deliberately excluded** from linkage.



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